Broadcast Technology UG1

Broadcast Workflow

Workflow

Effective workflow

- ensure that critical business processes can be tracked and optimised over time
- auditing, logging

- to seamlessly orchestrate the interactions between enterprise resources
    - storage
    - transcoding
    - network
    - etc

... and groups of people

Production to Delivery

PLANNING

- up to the point of shooting
- conventions and practices to be adopted at the start of the process

RUSHES MANAGEMENT

- capture and handling of content, on location or in a studio
- up to the point of rushes archive and management.

POST-PRODUCTION

- from the ‘ingest’ of material into the edit
- through to the completion of the master

DELIVERY

- production of masters for delivery to broadcasters, clients or audiences
- DPP technical and metadata standards for file-based delivery of completed programmes.

Sources, Technologies and Architectures

Contribution
- Dedicated links
- Satellite
- IP -based: low-res, high -res

Other Sources
- Archive
- News Agencies
- Stock footage
- Internet

If not networked, then probably …
- Tape
- Disk
- SSD
- Mobile, handheld devices

Metadata

Production
- Editing
- Colour grading
- Subtitles
- Metadata
- Metadata Input

- Archiving
- Asset Management
- (Predelivery – EPG .. etc)

Delivery
- Metadata

Distribution
- Terrestrial
- Satellite
- Cable
- Web: Live & VOD
- Mobile
- Metadata

User Content

Security, Asset Management

Typical Essences, Wrappers, Metadata, etc

ESSENCES – Audio, Video

- MPEG-2
- H.264 .. AVC-Intra
- Pro Res
- SHQ / NDI
- JPEG 2000
- DNxHD (VC-3)
- DNxRI (HR) (VC-3)
- DVCAM
- DVCPRO 25 / 50 .. HD
- AVI-DV
- MPEG – 1
- MPEG – 2 long GOP and I frame SD and HD
- MPEG PS, TS
- MPEG – 4
- AES3
  (S302M, S331M PCM, Dolby-E)
- AIFF
- Wave / BWF
- AAC

Codecs –

- Acquisition
- Mezzanine
- Distribution

METADATA

- Structural Metadata SMPTE 377M
- MPEG-7
- EBU-Core
- Descriptive Metadata Scheme – 1 (S380M)
- Custom Metadata Scheme Via XML
- AS11 1.1/DPP
- SMPTE 434 dictionaries

WRAPPERS

- AS11 1.1 / DPP
- XAVC
- Cinema DNG
- AVC-Ultra
- AVC-LonGOP (RDD 26:2014)
- AVC Proxy (RDD 25:2014)
- IMF
- MXF …
  - OPAtom and OP1a
  - OP1b; OP1C; Op2a-c; OP3a-c

Workflow – From Traditional to Next Gen TV

Recent Developments in the Media Industry (and specifically TV)

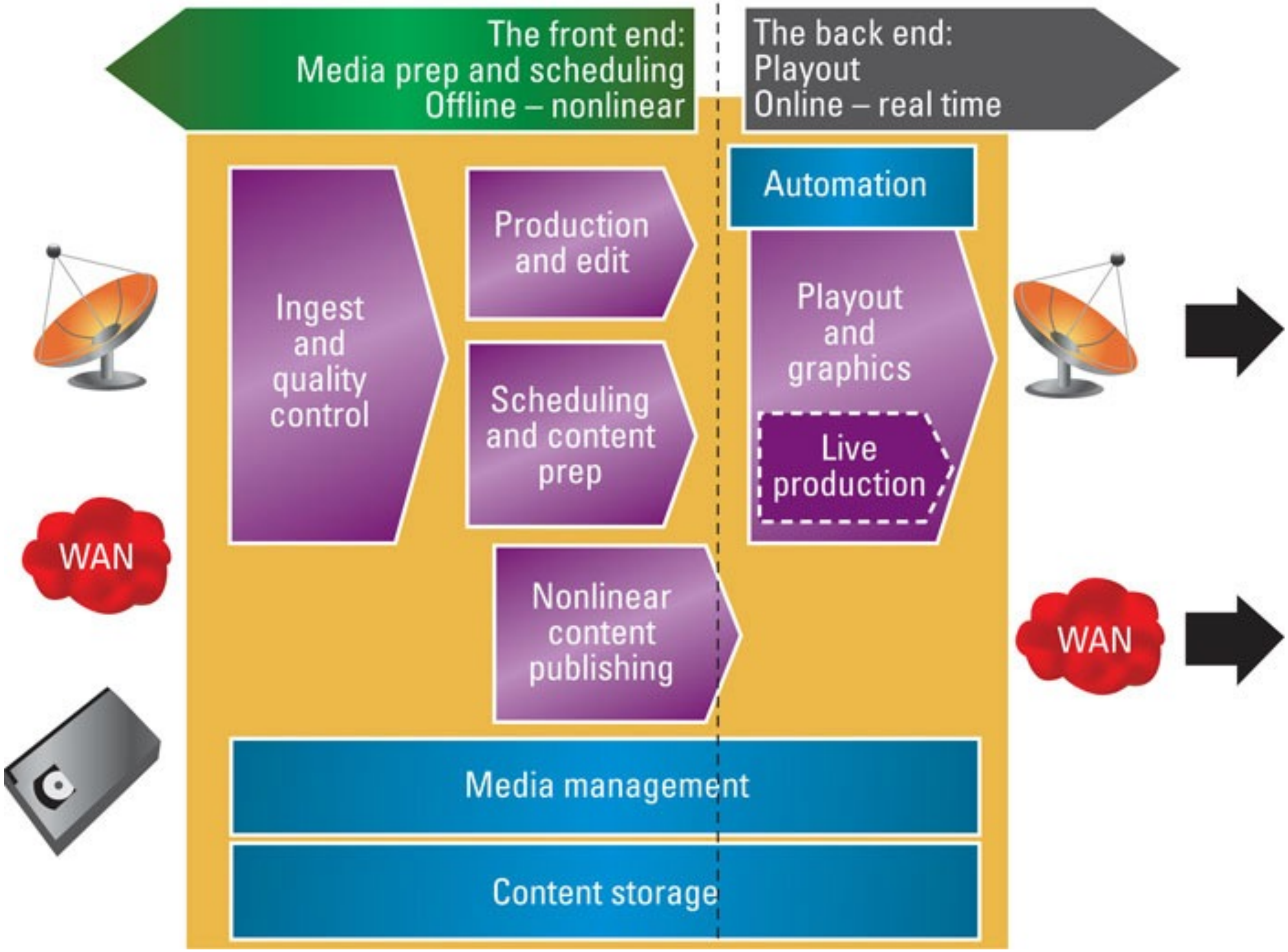
- **Main technologues**
    - UHD / 4K … 8K
    - HDR
    - video compression
    - storage
    - bandwidth

- **New services**
    - Digital Media / Multi Platform
    - Multi-screen
    - Interaction and personalisation
        - Multi-sourced inputs in real time
        - Multiple potential outcomes of programme
        - Ingest of social media for playout
            - Requirement for information and metadata

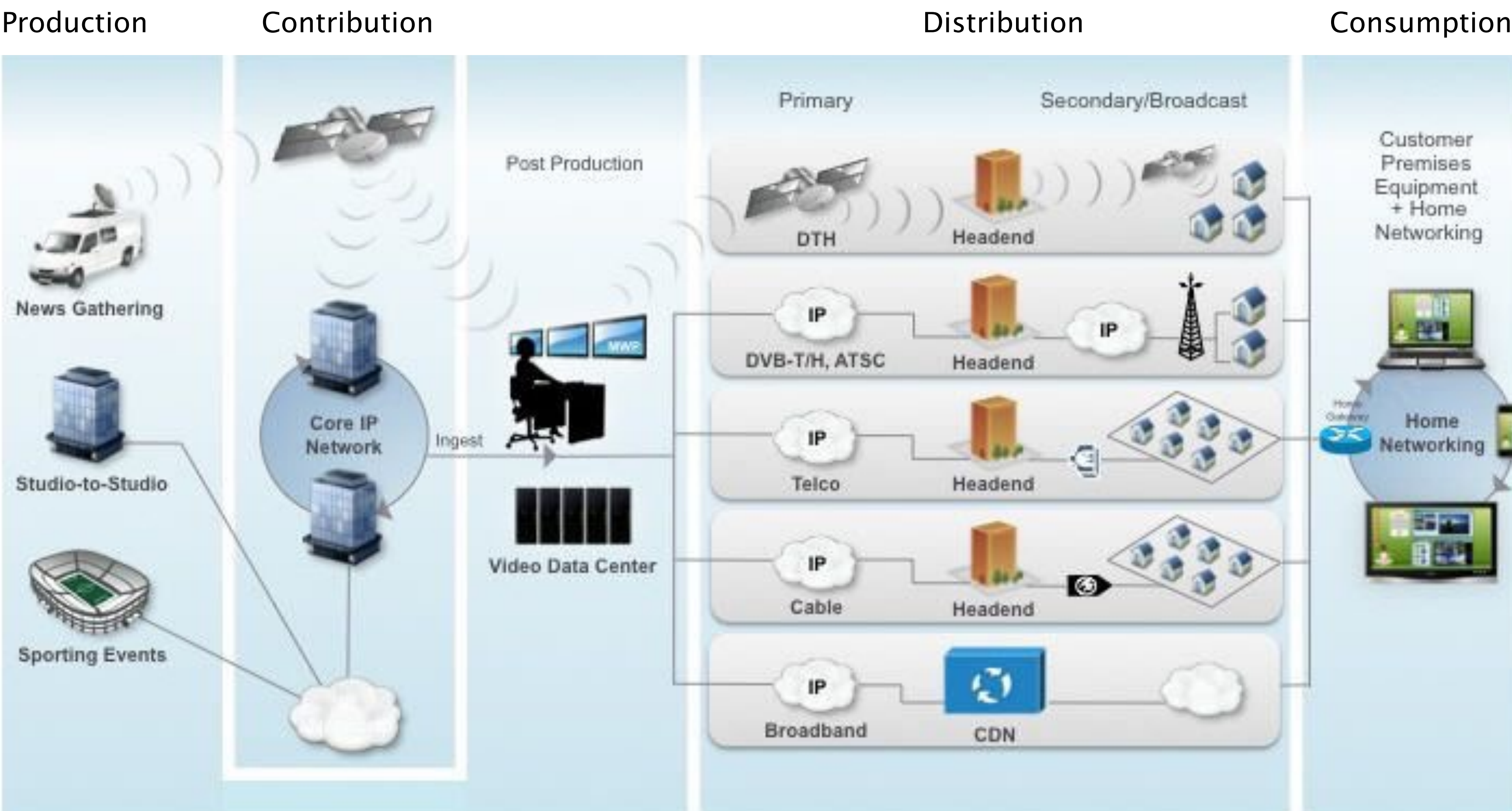
              ... to describe all constituent actors and objects in the programme
              ... their position and movement
              ... their relevance to the end user
              ... integration of content with other events and media
              ... Advertising and placement integrated in to the content
              ... 360º video
              ... VR / TV
              ... assisted TV, accessibility

Contribution and Distribution

Contribution and Distribution



Contribution and Distribution



http://www.cisco.com/c/en/us/solutions/service-provider/distribution/index.html#

Contribution and Distribution

Contribution (distinct from Distribution) – Transport of media content

– video, audio, metadata, and ancillary data – between professional users

- ingest
- capture and production activities

Key aspects for contribution

- Picture quality
- Latency
    - prime importance for contribution
    - end to end delay – encoding and decoding process
    - less important for distribution
        - distribution emphasis: maximizing picture within the available bit rate

- Synchronisation →
- Encryption →

http://cable.doit.wisc.edu/SMPTE/01-Compression%20for%20IPTV/Trow.TP.01-14-2008.pdf

Contribution Quality

Synchronisation

- close audio / video sync needed in production environment
    - tolerances –
        - −5 to +15ms = *"lip-sync"*
        - relaxed in contribution applications ~ −5 to +30ms)

c.f. **<u>Distribution</u> applications**

- tolerances significantly relaxed
- lip-sync acceptable −40 to +60ms limits

Encryption

- contribution
    - limited number of encryption standards
- distribution
    - wide array of encryption and entitlement schemes being commonly used

Contribution Quality

According to DVB Commercial Module:

- Quantified as equivalent to uncompressed or very mildly compressed source material
    - assumed that the material is full frame rate
    - usually 4:2:2 chroma-subsampling sampling

Compare to:

- Broadcast Quality
    - equivalent to Satellite, Terrestrial and Cable broadcast
    - full frame rate but at a quality not assumed appropriate for contribution applications

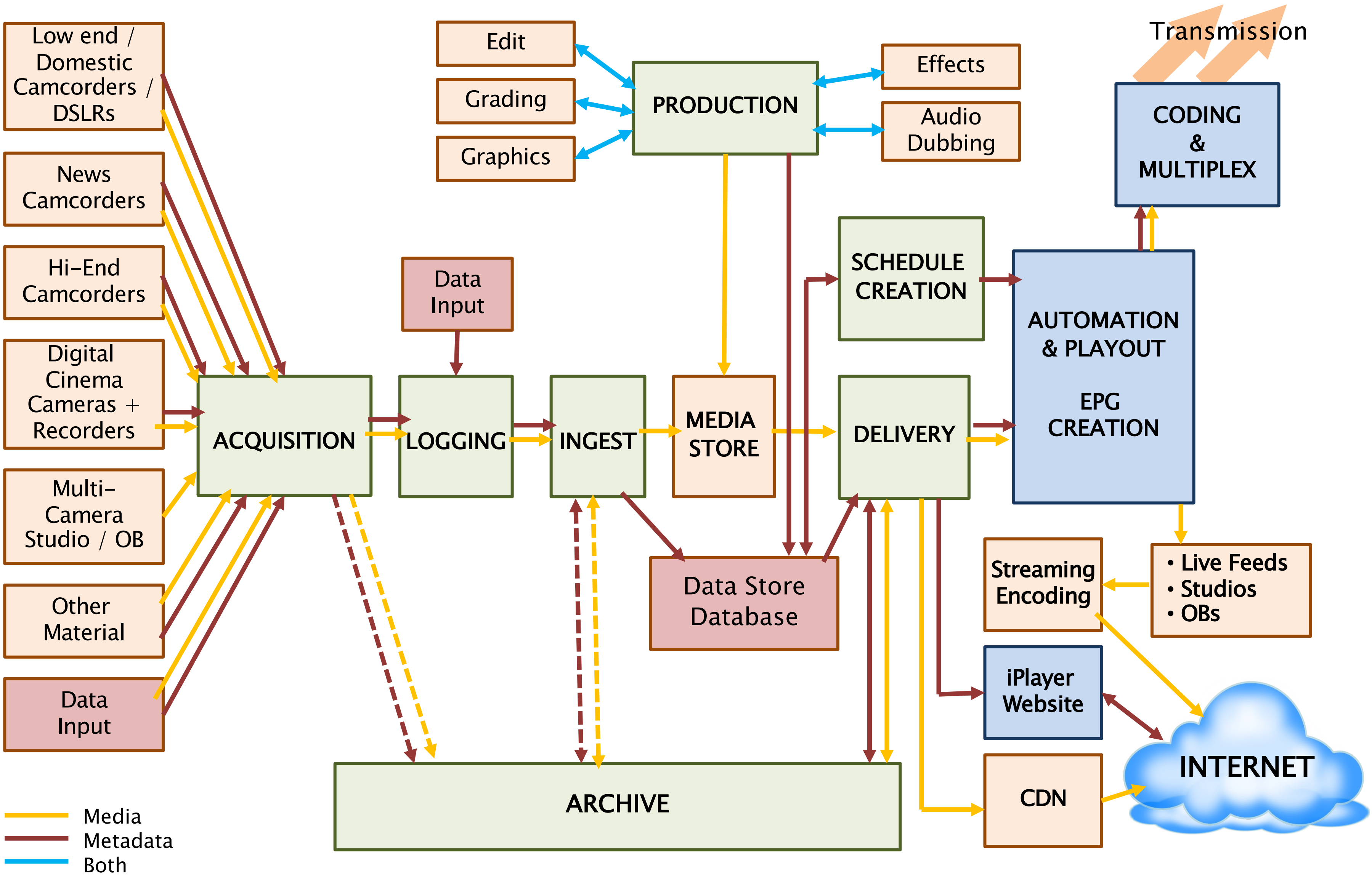
- Good Quality
    - commercially acceptable quality
    - not considered acceptable for contribution except as a last resort
    - or for fast response news applications
    - not necessarily full frame rate

- Acceptable Quality
    - commercially acceptable for low data rate applications

- Mobile Phone Quality
    - equivalent to CIF and QCIF resolutions (phones, etc)
    - (but more HD phones now)

CIF – Common Intermediate Format – resolution of 352 × 288
QCIF – Quarter CIF

Broadcast and Production Workflow (BBC Schematic)



Broadcast and Production Workflow (BBC Schematic)

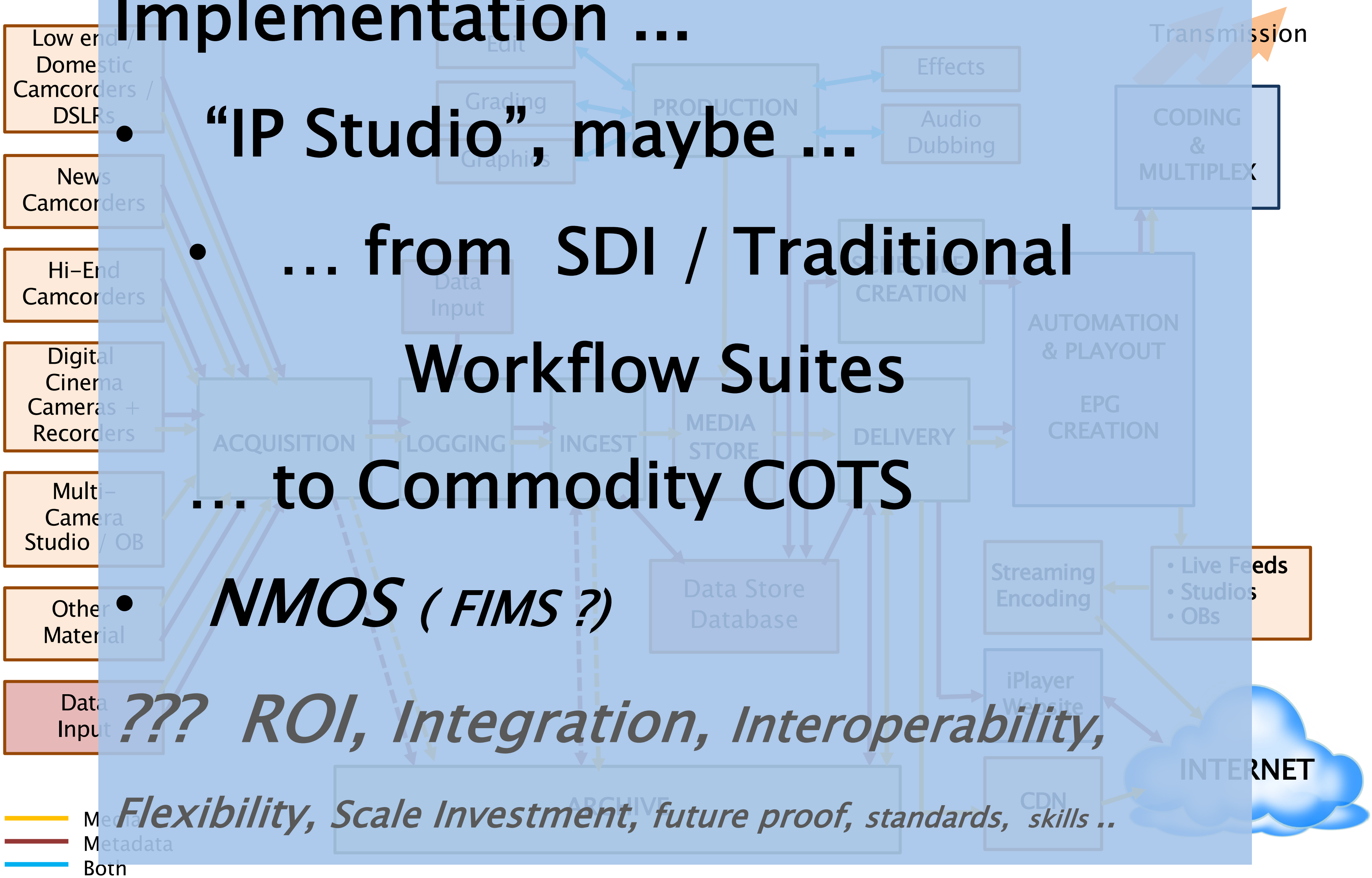
Implementation ...

- "IP Studio", maybe ...
  - ... from SDI / Traditional Workflow Suites

... to Commodity COTS

- *NMOS (FIMS ?)*

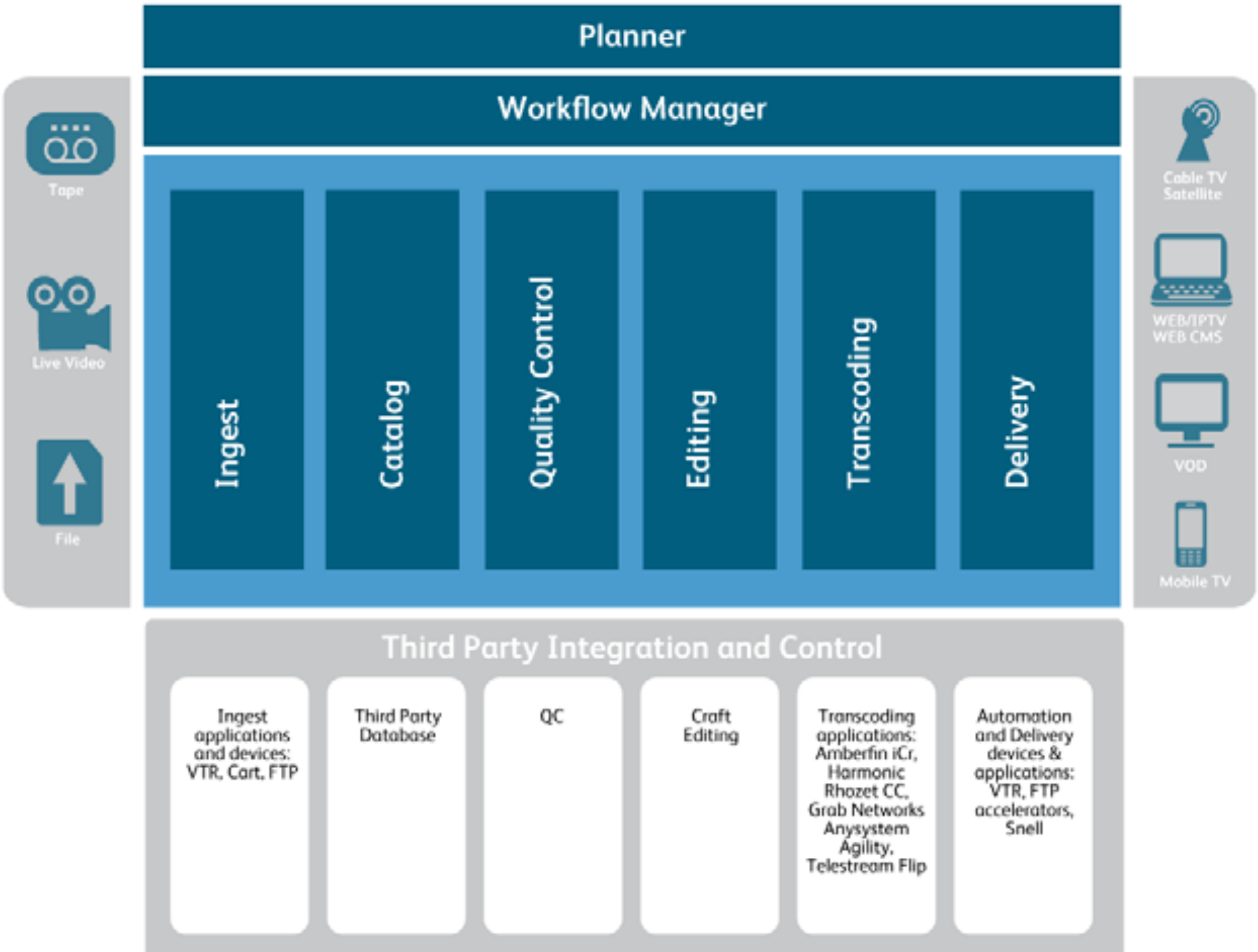
??? ROI, Integration, Interoperability, Flexibility, Scale Investment, future proof, standards, skills ..



Overview of many / some of Workflow Processes

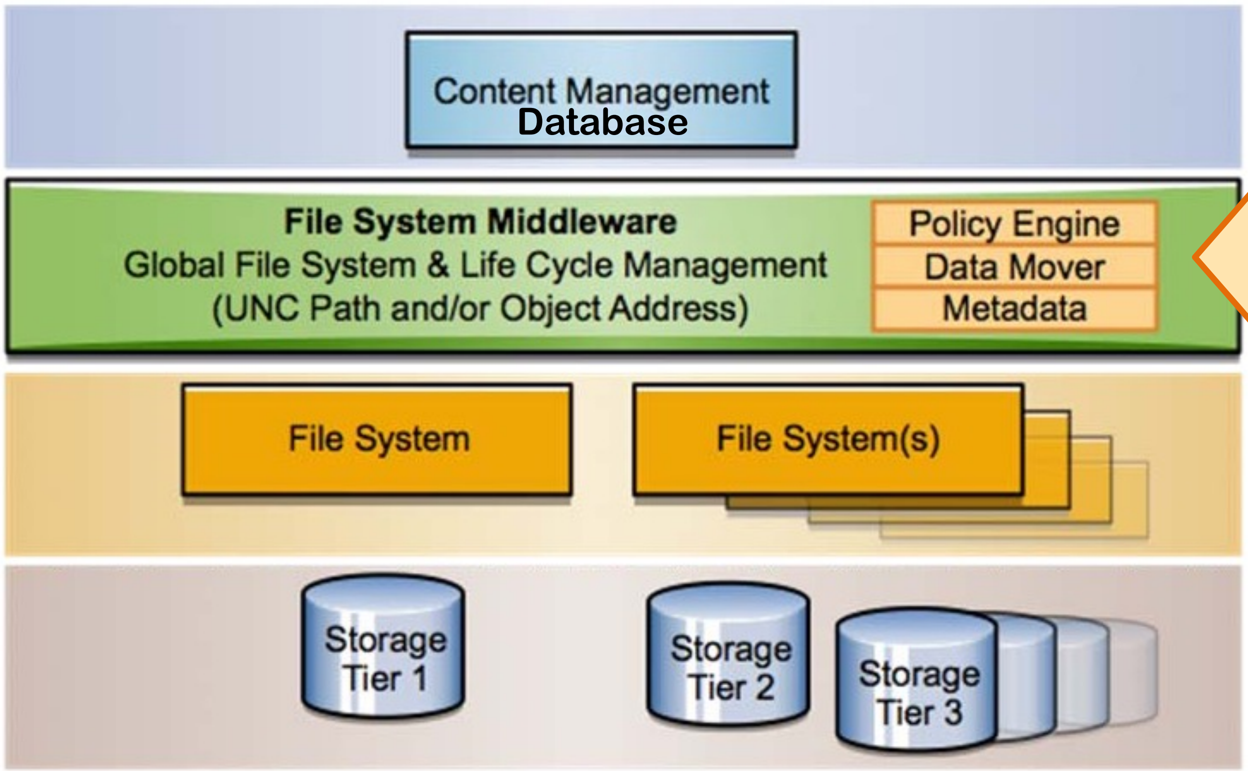
Workflow Processes

Typical – there are many variations ...
... and automated:



http://www.snellgroup.com/documents/brochures/automation-and-media-management/Momentum.pdf

MEDIA ASSET MANAGEMENT



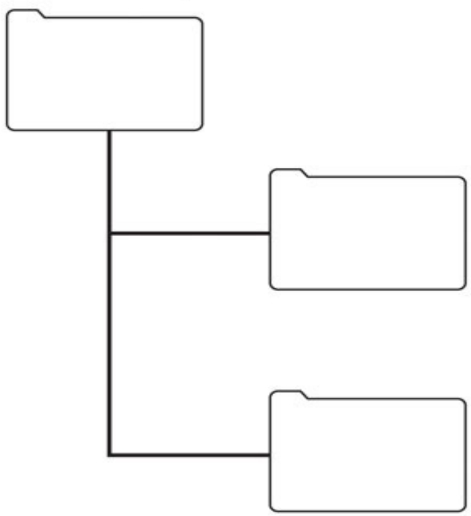
BPM: Business Process Management ("Workflow Engine")

Typical MAM File System Approach

Preferred approach – file-folder structure

- accessible via a standard network protocol (rather than a proprietary interface between the storage and the asset management system)

Archival file system
software layer



File 1: DISK H1 … written today
File 2: LTO L3
File 3: DISK H4 … being copied to LTO L2
File 4: LTO L2
File 5: DISK H3 … written last hour

File 1: LTO L7
File 2: LTO L5
File 3: DISK H2
File 4: LTO L1 being copied back to
disk (due to recent read attempt)

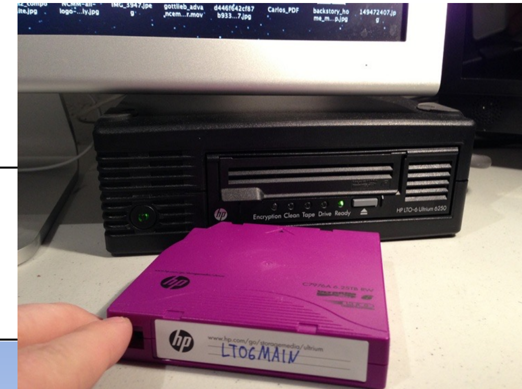
+ access to other cloud storages
(own, third party)

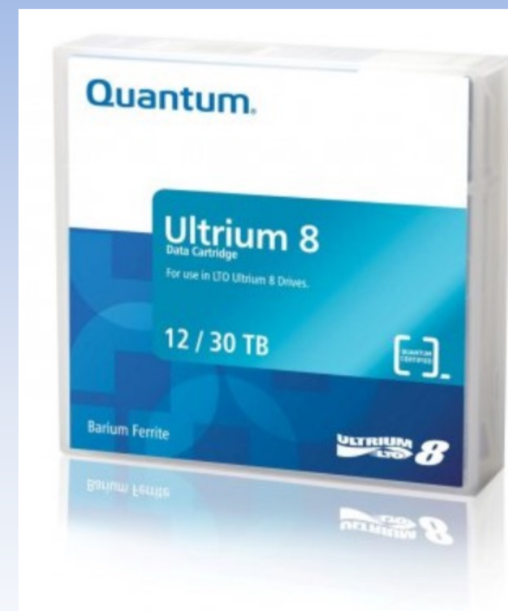
Linear Tape Open – to LTO 10

General specifications

Tape type	Raw capacity (compressed capacity @ 2.5:1)	Max Uncompressed Speed MB/s (note bytes)	WORM	Released
LTO-1	100 GB	20	No	2000
LTO-2	200 GB	40	No	2003
LTO-3	400 GB	80	Yes	2005
LTO-4	800 GB	120	Yes	2007
LTO-5	1.5 TB	140	Yes	2010
LTO-6	2.5 TB 6.25 TB	160	Yes	2012
LTO-7	6.4 TB 16 TB	315	Yes	2015
LTO-8	12.0 TB 30 TB	472 (upto 900Mb/s)	Yes	Now
LTO-9	26 TB 65.2 TB	708	Planned	TBA
LTO-10	48 TB 120 TB	1100	Planned	TBA

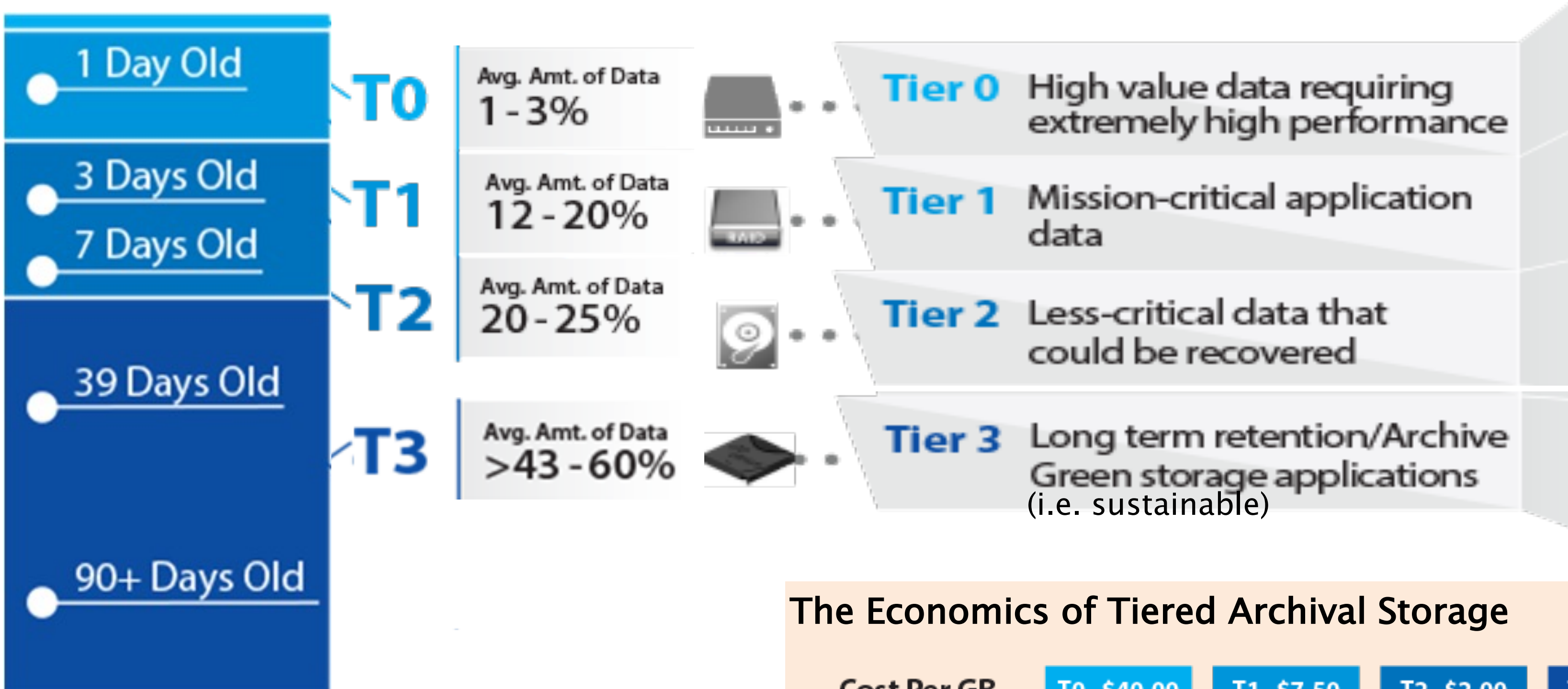
Note comparative native / compressed capacities

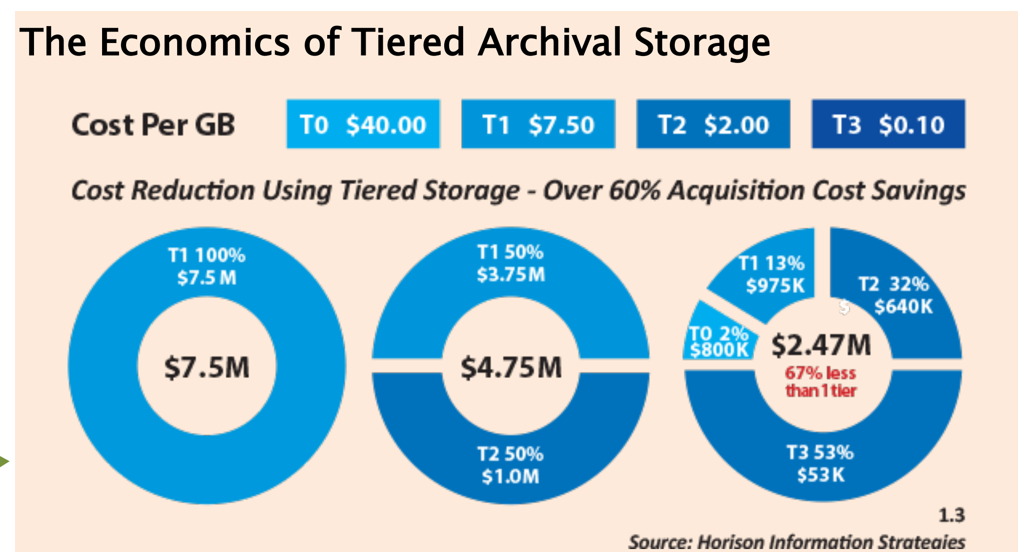




(2014 …)
Sony Tape @ 148Gb/in2
… 185TB per cartridge

Tiered Archival Storage





Tiered storage utilizing tape

- reducing power
- cooling
- space
- media and administration costs
- efficient access via LTFS indexing
- infrastructure cost reduction

http://perpetualstorage.com/need-right-data-right-place/ [as at 2013]

Cloud Archival

Storage of an organization's media assets:

- *original content*
- *EDLs*
- *finished masters*
- *copies*
- *versions*
- *releases, etc*

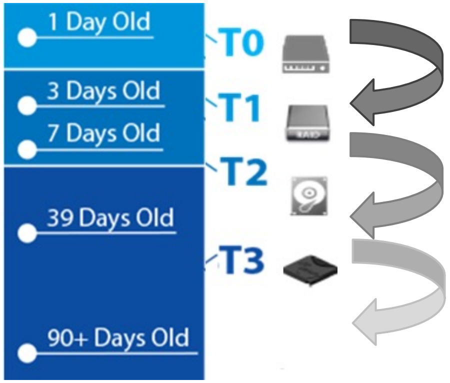
Cloud-based storage being used for

- *ingest and content collection*
- *playout from the cloud*
- *processing in a virtual environment*

.. so what is the need (and issues) **for** ***storage in the cloud*** ?

Cloud concerns

- storage alternatives
- tier-based migration
- automated management



Storage alternatives

- Backup or archive
- *On-prem* or in *co-lo*
- Private, hybrid or public cloud

MXF – Material eXchange Format

- wrapper
- interchange of finished / almost finished audiovisual material
- + metadata

AAF – Advanced Authoring Format

- wrapper – essence + metadata
- post-production stages

Advanced Authoring Format (AAF)

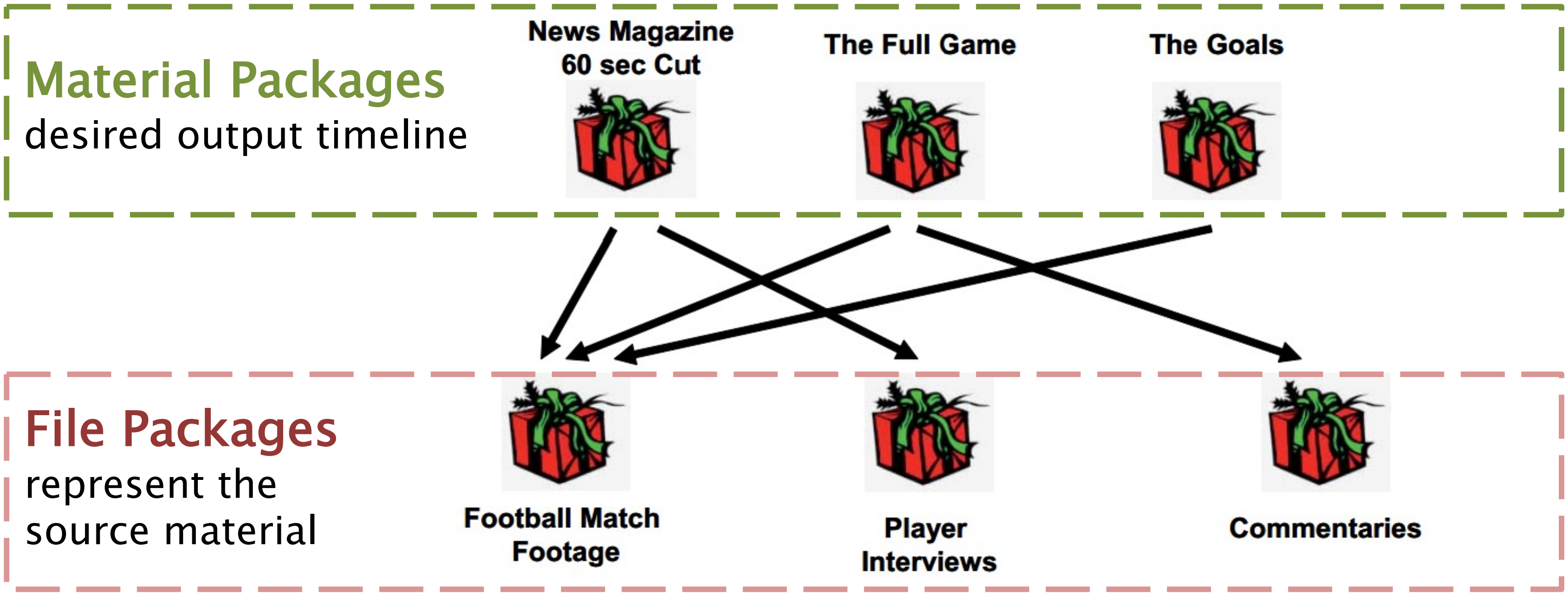
- digital nonlinear post production and authoring
- "Super-EDL"

- supports essence + associated metadata
  - provide cross-platform interchange

    of metadata or program content
  - track history of program content

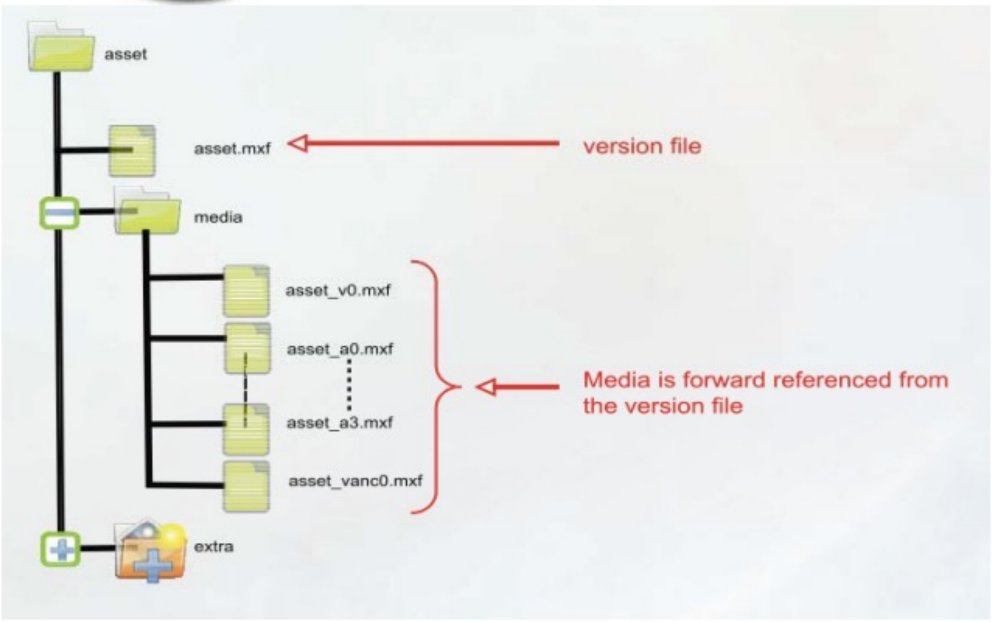
    from source elements through final production
  - content to be described as a media object

MXF – Material eXchange Format



- e.g. clip composed from a pair of recordings
  - MXF file has –
    - 2 File Packages and
    - 1 Material Package (instructs decoder how to play the packages)

VERSIONING



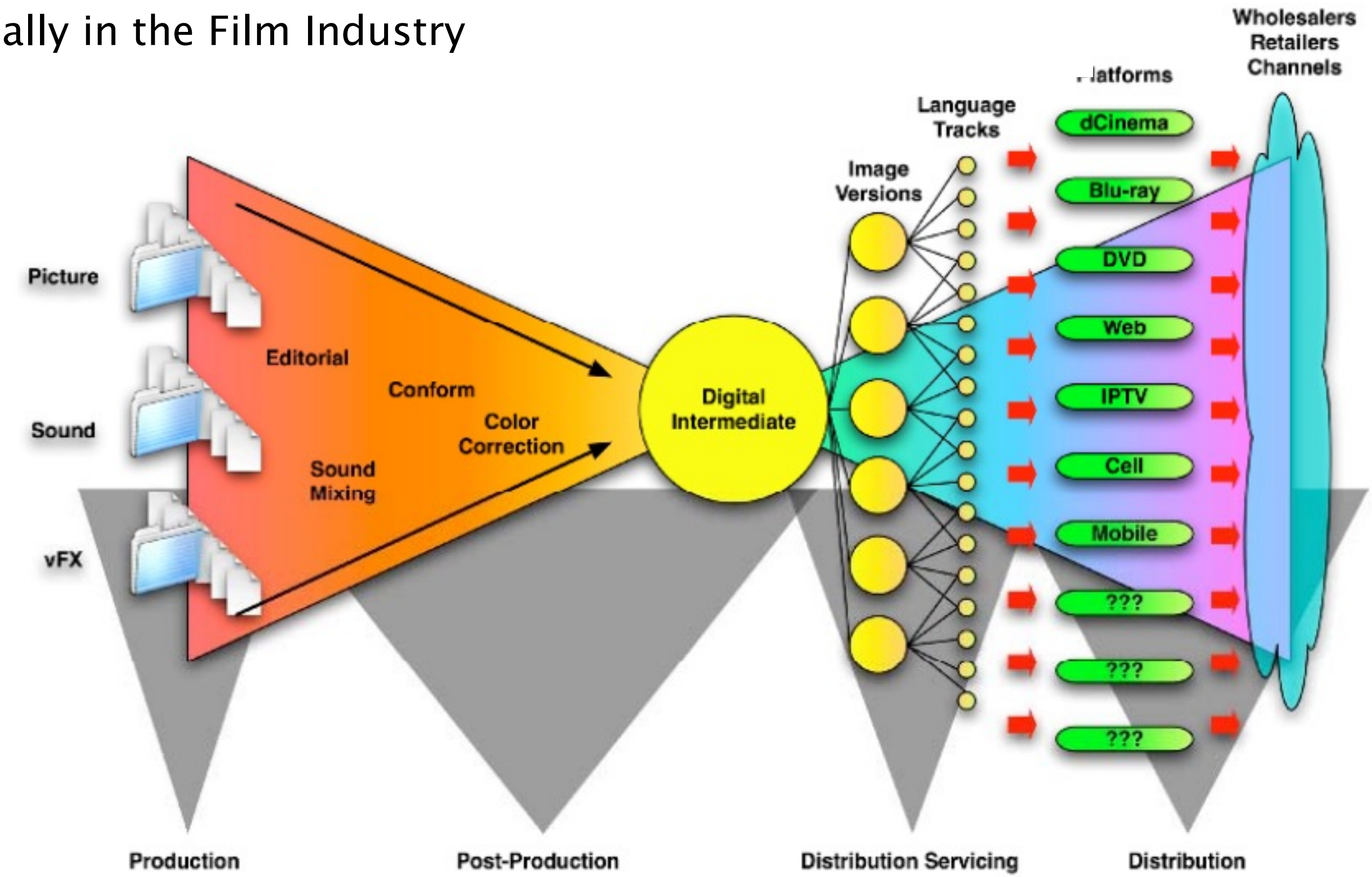
AS-02 – typically in TV broadcast

Versioning - IMF

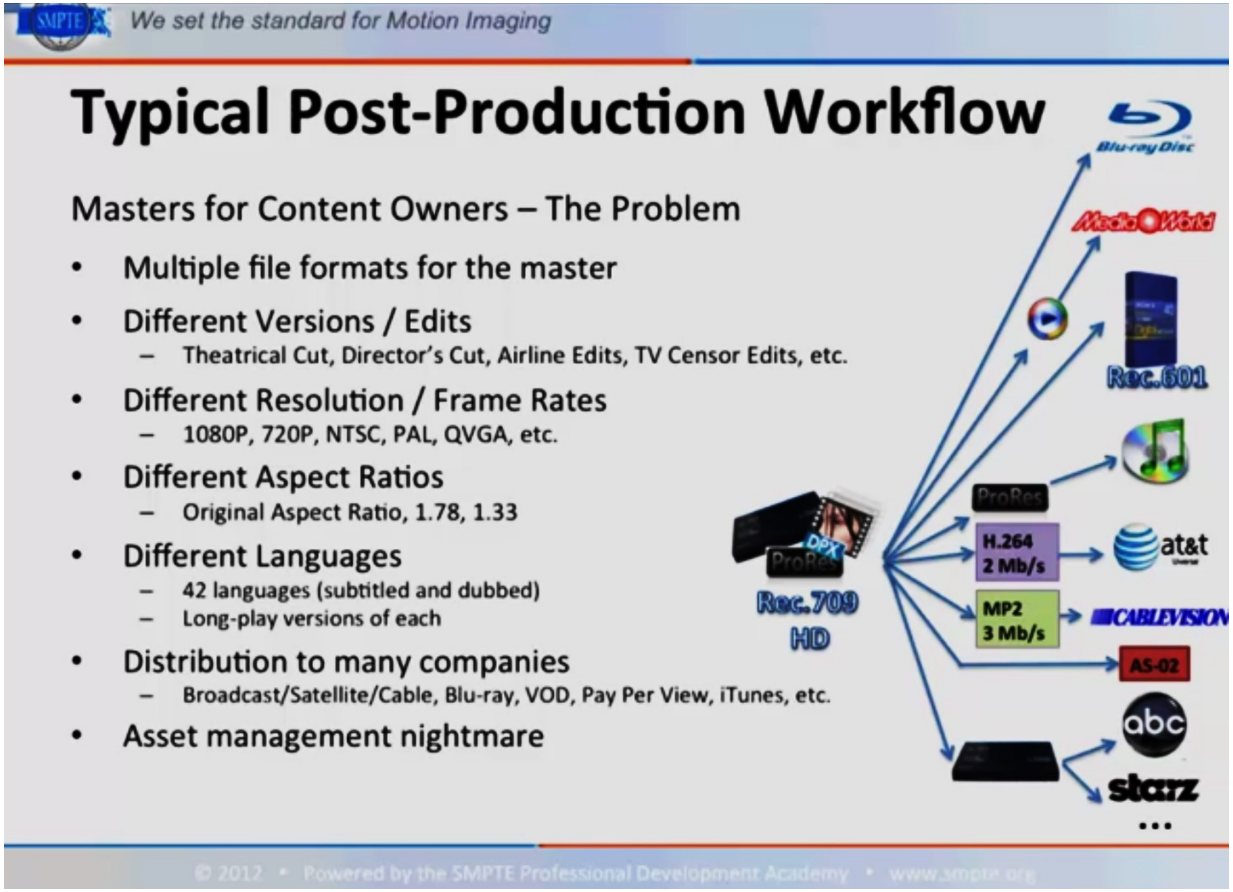
IMF - Interoperable Master Format

High Quality Multi Version Masters for Studio Applications

Typically in the Film Industry



IMF – Interoperable Master Format



https://www.smpte.org/PDA_On-demand/IMF (1h 09m)

IP Workflows SMPTE 2022, 2110

IP Workflows

Why IP is replacing SDI*

- Universal connectivity (broadcast and other industries)
- Lower budgets, higher quality requirements
- Production companies scaling:
    - IP allows best production talent remains in a central location
    - Receive camera feeds from remote locations
- Democratise the industry
    - single producer can have many camera inputs from anywhere in the world

** although SDI is still in development*

How is IP Television Routing Different from SDI ?

In SDI Routing, all routing in the SDI crosspoint frame

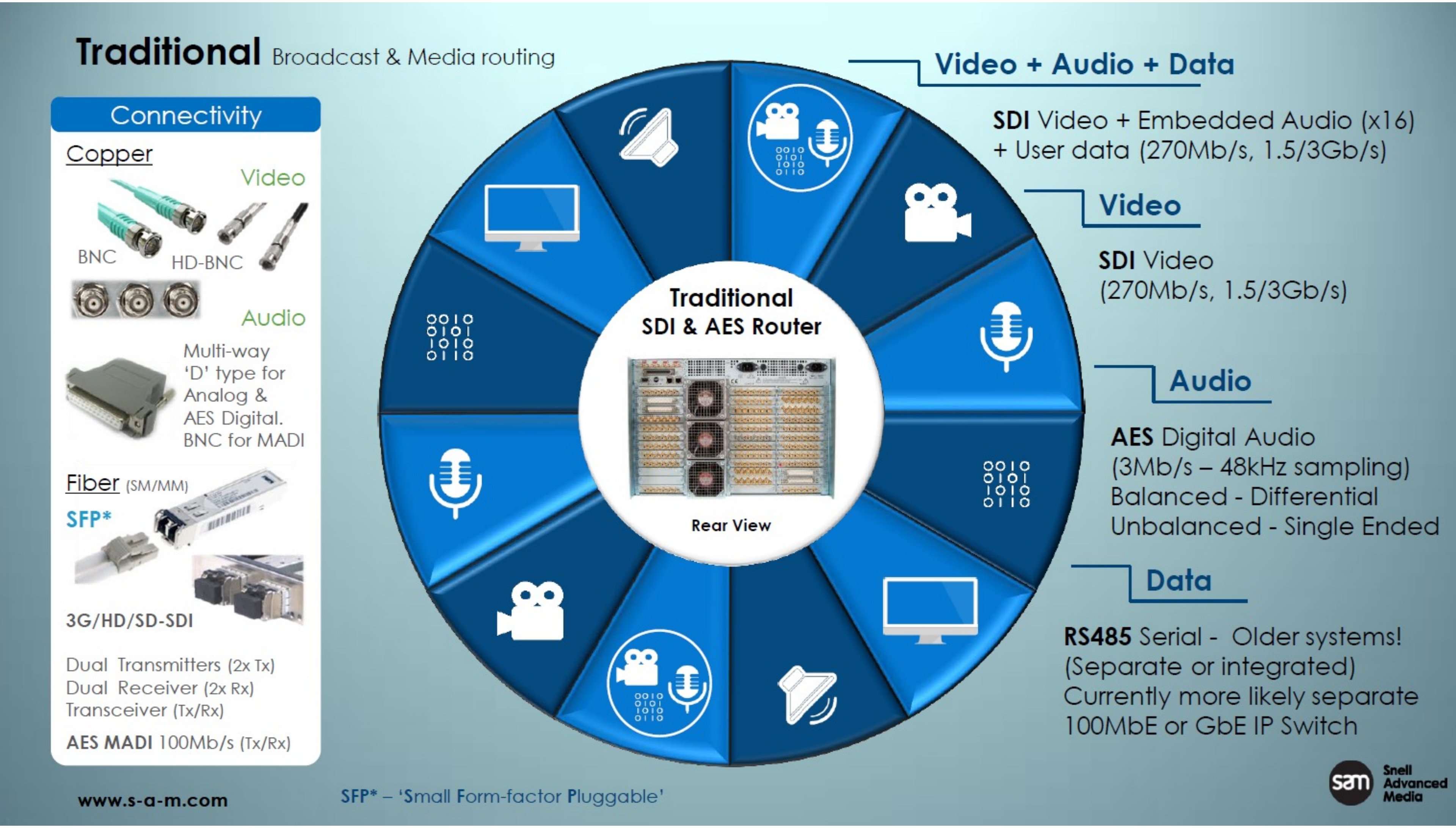
- The Control System tells the crosspointwhat to switch where
- The Crosspoint Router switches everything internally
- Receivers just receive all data sent to them

In IP Routing, the Receiver is involved …

- Control System tells the Receiver:
  - What Multicast Group & Port# to switch to (Main and Protect)
  - Technical Metadata of the new stream
- The Receiver is responsible for how it switches/transitions to it
- The Receiver asks the network for the new signal
- The network provides the new signal through packet forwarding logic

https://www.smpte.org/sites/default/files/section-files/GVExpo2018%20SMPTE%2004%20-%20Scalable%20Facilities%20with%20ST%202110%20-%20Mailhot.pdf

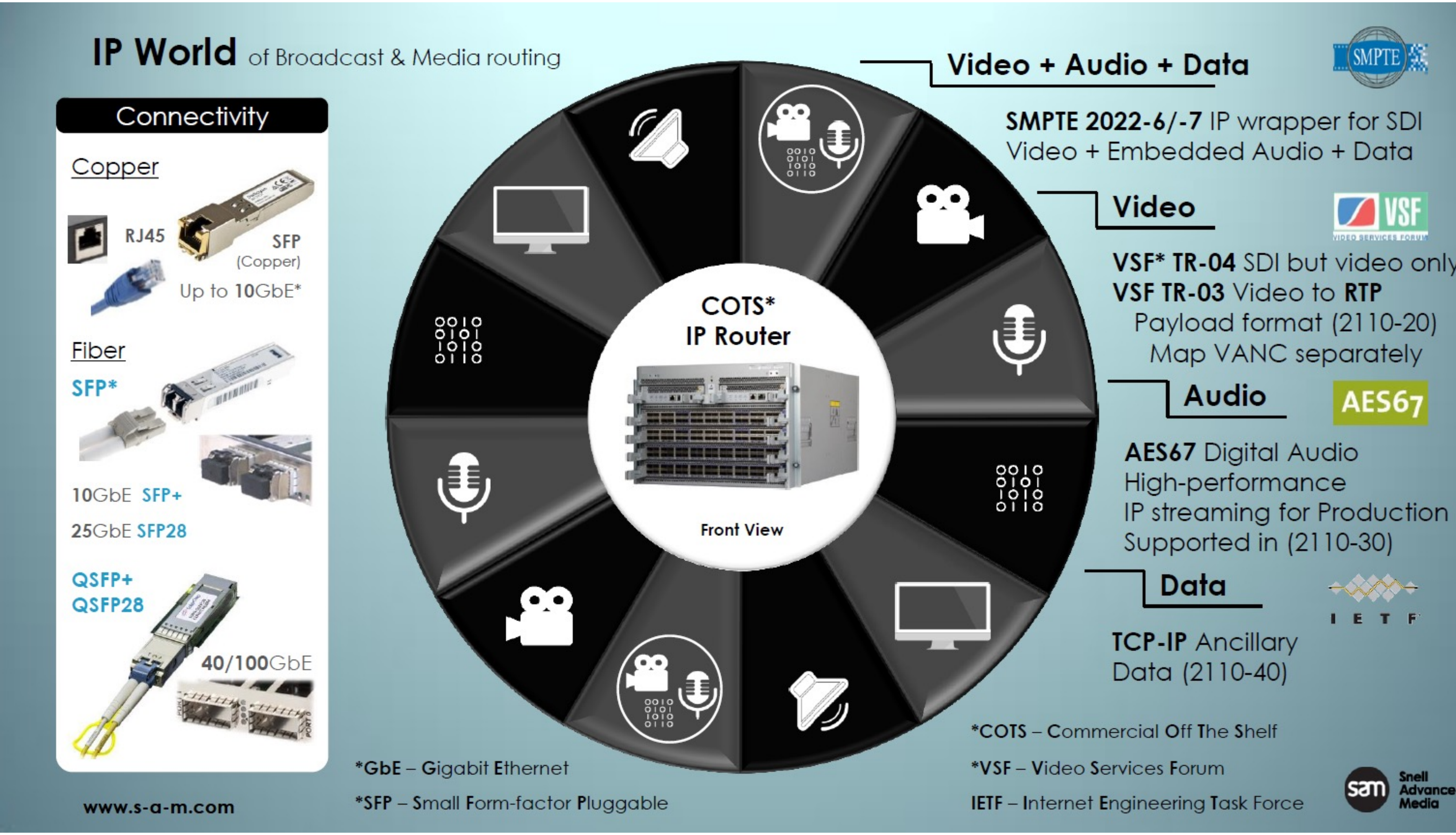
Traditional Broadcast and Media Routing



Snell Advanced Media

Slide from Phil Myers, Snell Advanced Media, SMPTE presentation, BCU. 2017

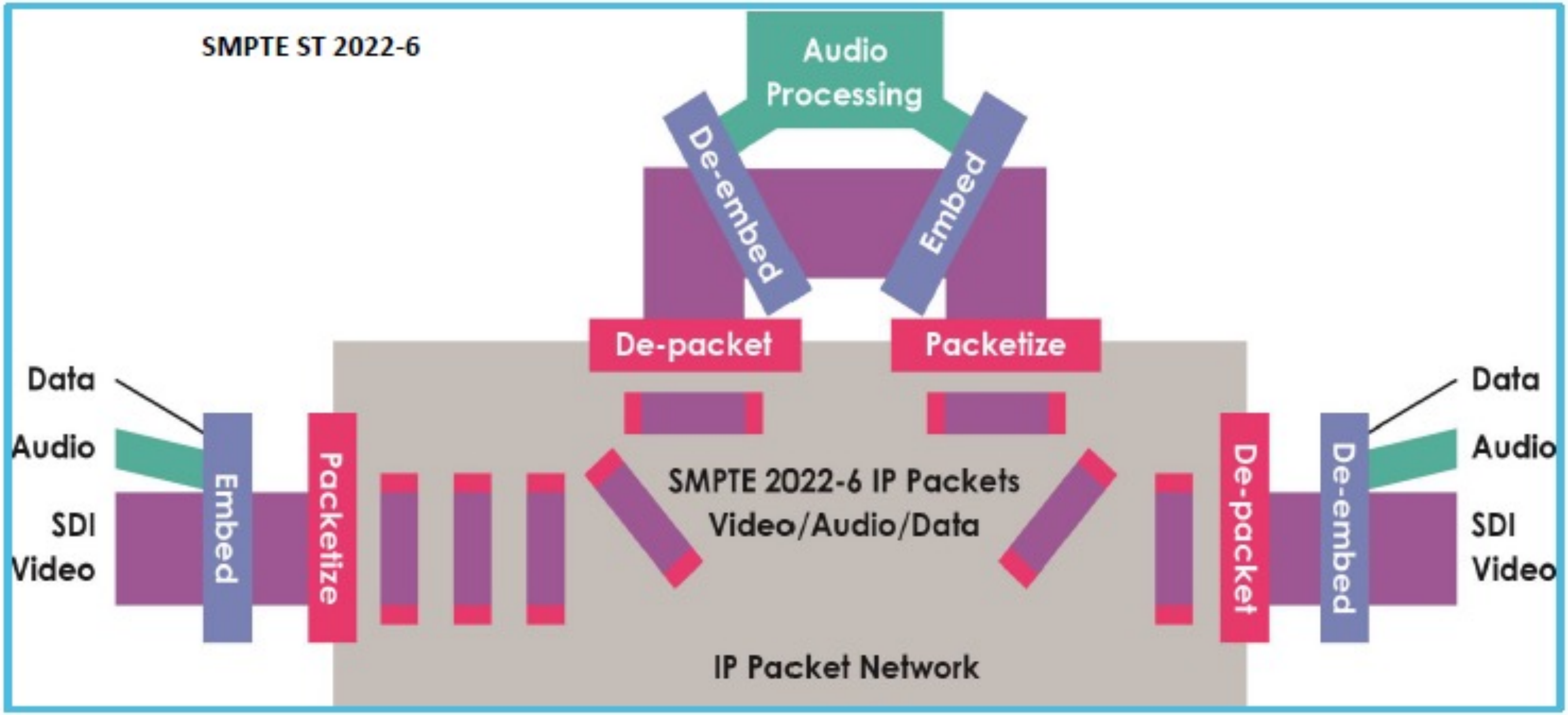
IP World of Broadcast and Media Routing



Snell Advanced Media

Slide from Phil Myers, Snell Advanced Media, SMPTE presentation, BCU. 2017

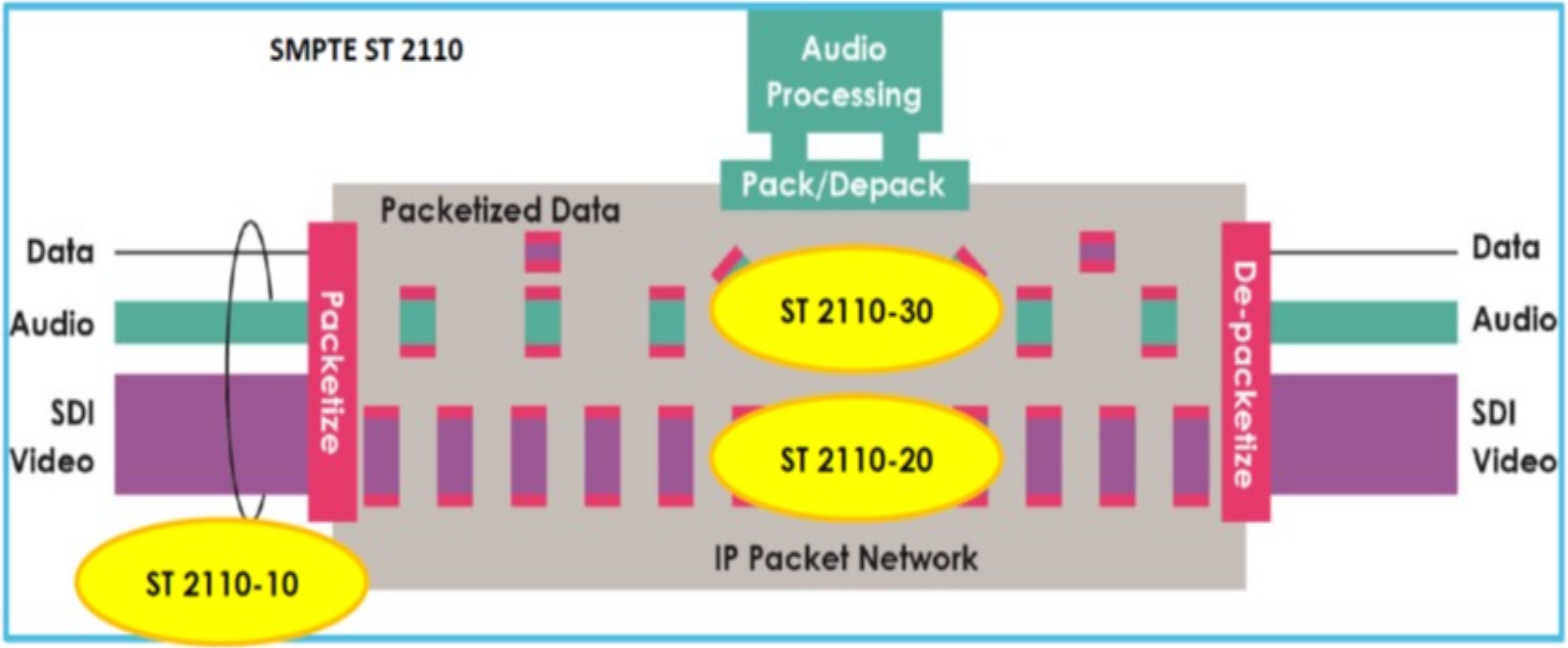
SMPTE ST 2022-6 (Multiplexed Streams)



Snell Advanced Media

Slide from Phil Myers, Snell Advanced Media, SMPTE presentation, BCU. 2017

SMPTE ST 2110 – Essence Streams

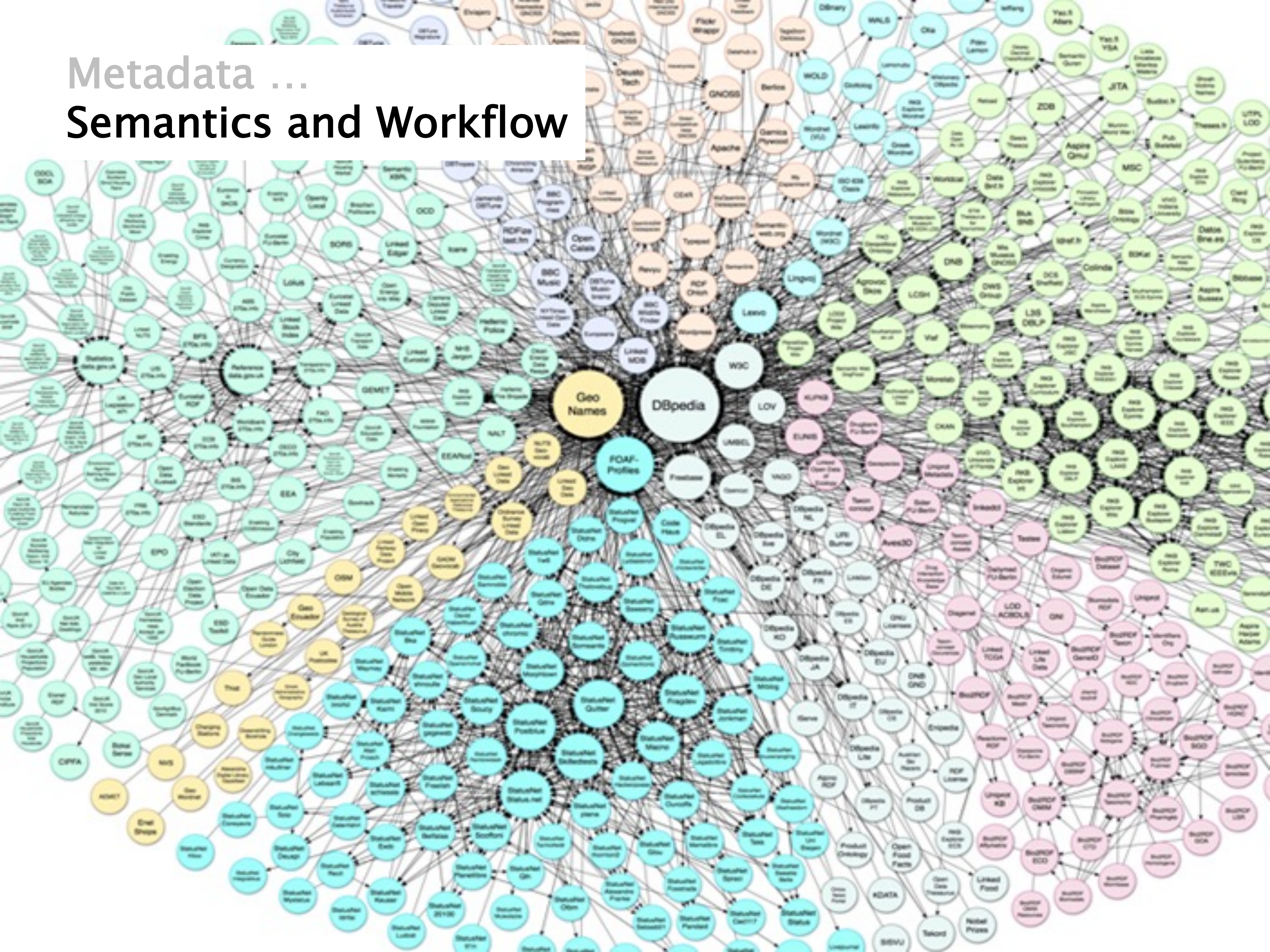


Slide from Phil Myers, Snell Advanced Media, SMPTE presentation, BCU. 2017

Workflow
.. next ...

Metadata …

Semantics and Workflow



Service Based Workflow – SOA / FIMS / Microservices

Having split workflow into services …

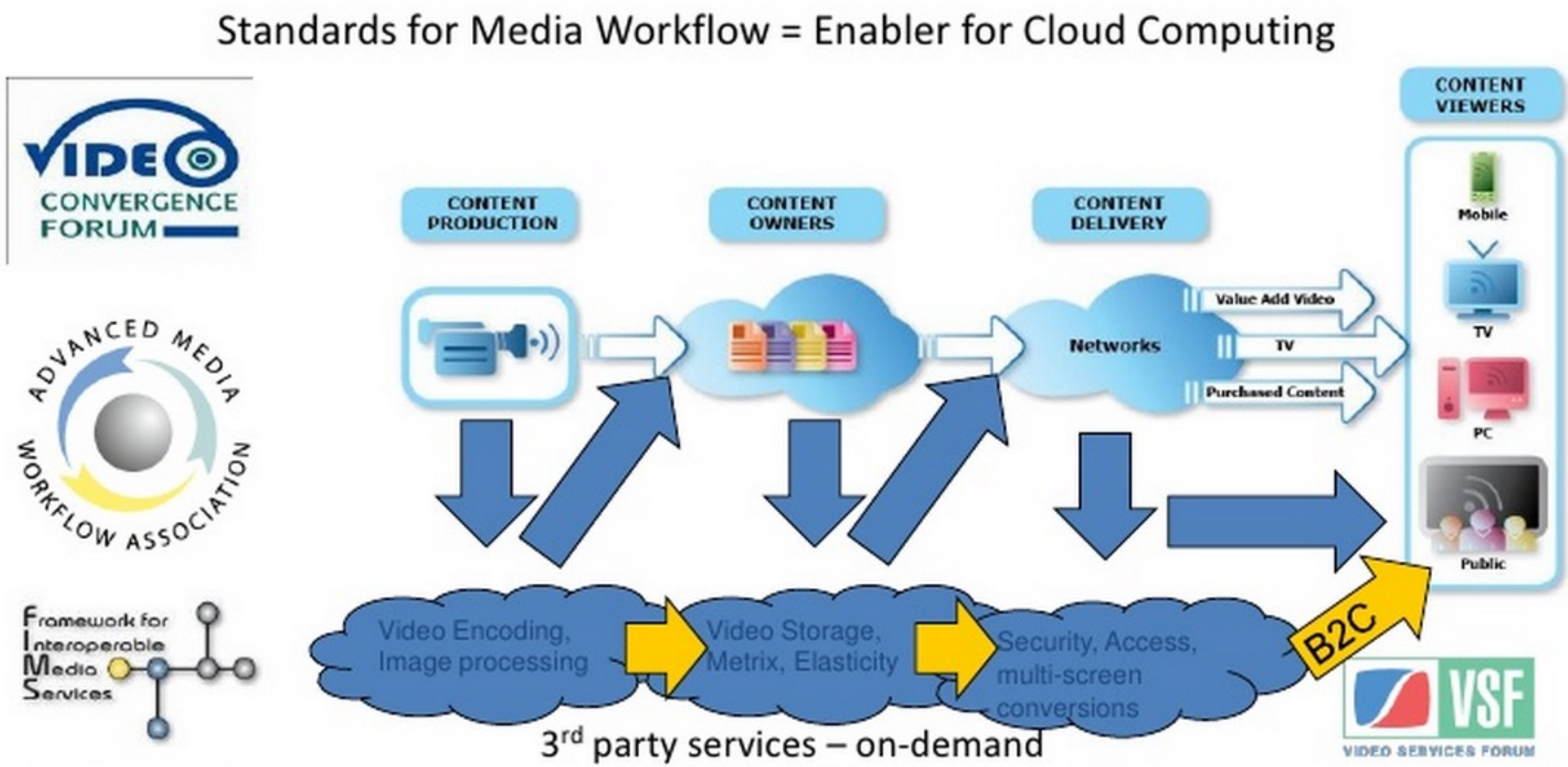
- Can outsource it and grow utilisation of third part sevices
- … into the cloud ..

- Could be another operational model for MXF?

- Broadcaster flexibility in whether / or not to invest in infrastructure

- Microservices

Service Based Workflow … cloud based

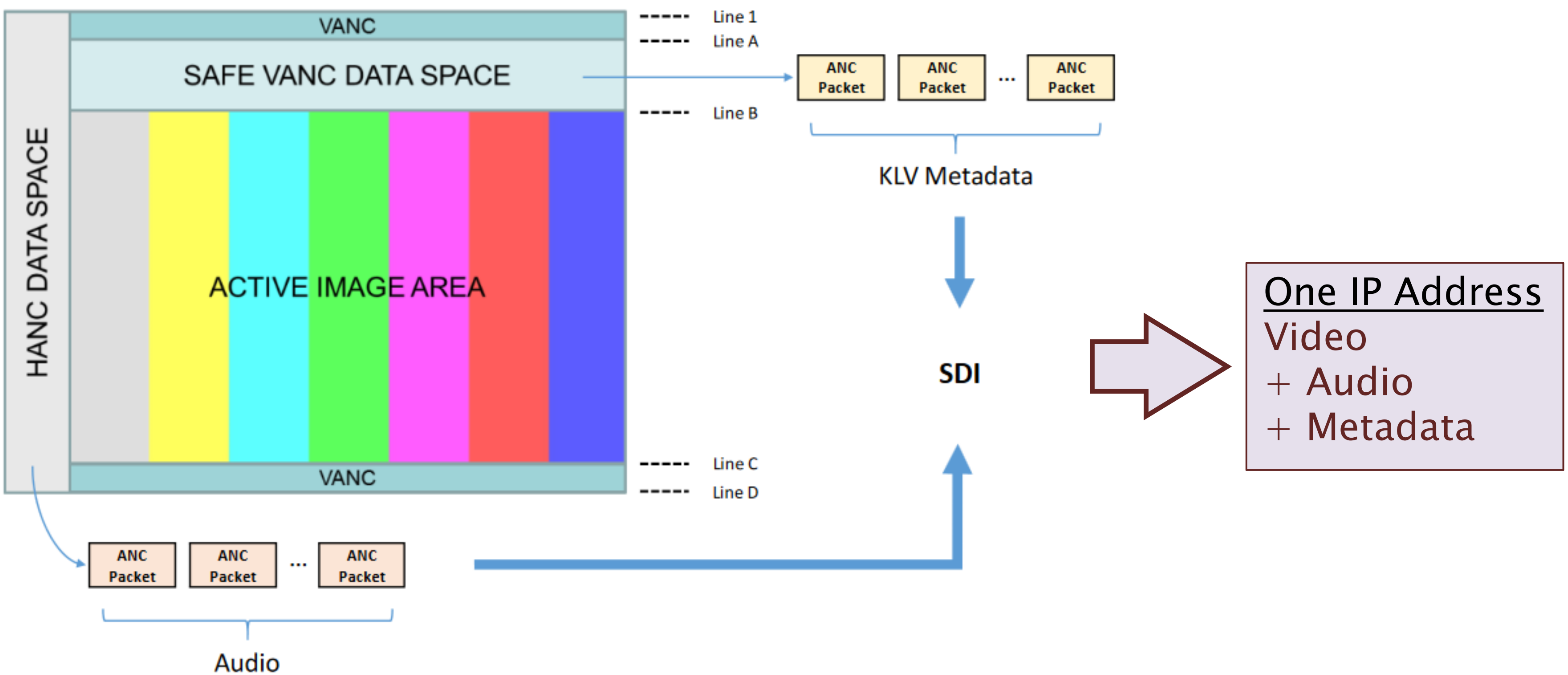


MJDrouin, VSF

SMPTE ST 2022-6 – IP Packetisation of SDI Raster

<u>Bundled (Audio, Video, Metadata together)</u>

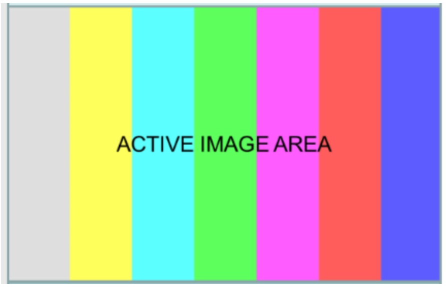
- Audio/Video/Metadata/Sync travel coherently
- Requires extra work to “unpack” separate essences



ST 2110 – Essence-Based Approach

Essence Based (Audio, Video, Metadata separate)

- Suited for Studio/Production workflows
- Individual essence kept in sync using PTP timing
- No blanking!



IP Packetisation of Active Video

IETF RFC 4175

IP Address #1
Video

IP Packetisation of Audio Channels

AES-67

IP Address #2
Audio

IP Packetisation of ANC Data

IETF ANC291

IP Address #3
Metadata

The business case for delivery and control standards in program workflows – Bruce Devlin, 2014

A historical perspective leading to Digital Delivery, but a highly relevant argument of various interoperable developments across the industry

https://youtu.be/K3dn5gQYyw4

(at 11m15 there's a section on FIMS: FIMS is becoming less relevant, but the underlying issue is still important – note my comments in class)